

Probabilistic Role Coordination in Multi-Agent Systems

Using Bayesian Inference with Memo

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Agenda

1. Introduction
2. Part 1: Basic Game
3. Part 2: Stateful Extensions
4. Part 3: Multi-Factor Experiments
5. Conclusions

Introduction



The Central Question

**How do agents infer their role within an institution,
and how does this shape their actions?**

The Challenge:

- Multiple agents must coordinate without central authority
- Each has specialized capabilities but uncertain about their role
- Must infer roles from observing each other's actions

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Our Approach:

- Model agents using Bayesian inference over role assignments
- Start simple, then add complexity (states, capabilities, difficulty)
- Systematically analyze what makes coordination succeed

Key Concepts

Institutional Roles

- Roles define behavioral expectations (Fighter, Tank, Healer)
- Actions are probabilistic given roles
- Complementary roles improve team performance

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- Agents observe actions and update beliefs (Bayesian)
- Converge to shared understanding through iteration

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The Memo Framework

- Probabilistic programming for Bayesian inference
- Express generative models and perform inference
- Natural fit for role coordination problems

Part 1: Basic Game

Game Setup

Three Roles:

- **Fighter:** Specializes in attacking
- **Tank:** Specializes in defense
- **Healer:** Specializes in support

Three Actions:

- **Attack:** Deal damage to enemy
- **Defend:** Protect team
- **Heal:** Restore team health

Game Setup

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Role Policies: Each role has a preferred action distribution

```
ROLE_ACTION_PROBS = [  
    [0.70, 0.25, 0.05], # Fighter: mostly attacks  
    [0.25, 0.70, 0.05], # Tank: mostly defends  
    [0.25, 0.05, 0.70], # Healer: mostly heals  
]
```

Bayesian Inference with Memo

Inference Process:

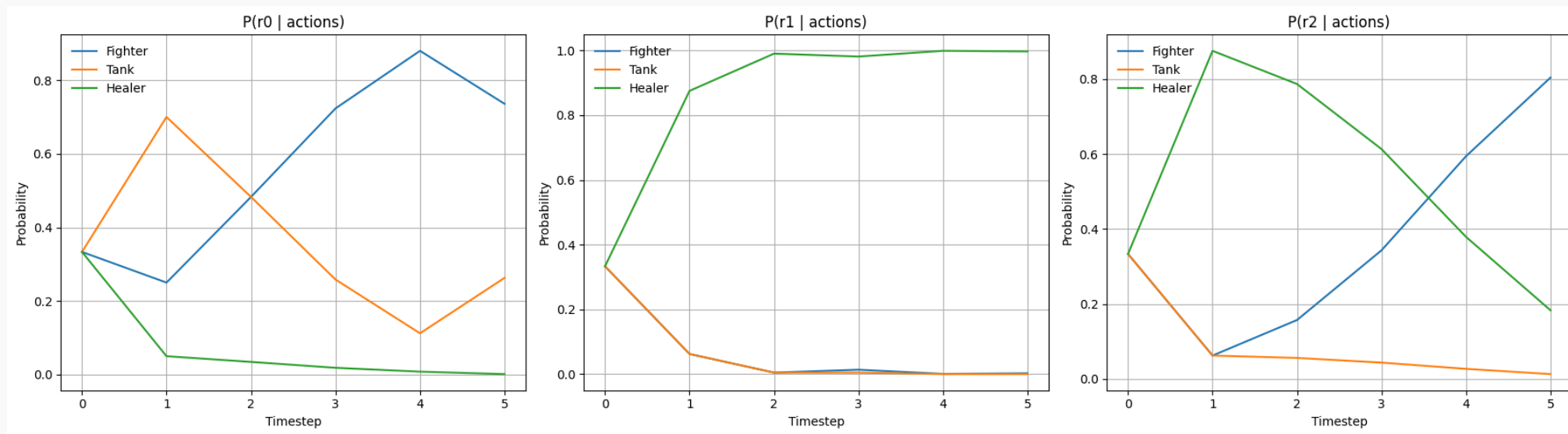
1. Prior: Beliefs about roles
2. Likelihood: Role → Action
3. Evidence: Observe actions
4. Posterior: Update beliefs

```
@memo
def role_inference[r0, r1, r2](
    role_prior, obs_a0, obs_a1,
    obs_a2):
    observer: knows(r0, r1, r2)
    observer: thinks[
        team: assigned(r0, r1, r2,
            wpp=role_prior[r0,r1,r2]),
        team: chooses(a0,
            wpp=role_policy(r0, a0)),
        team: chooses(a1,
            wpp=role_policy(r1, a1)),
        team: chooses(a2,
            wpp=role_policy(r2, a2))
    ]
```

```
observer: observes_that  
    [team.a0 == obs_a0]  
observer: observes_that  
    [team.a1 == obs_a1]  
observer: observes_that  
    [team.a2 == obs_a2]  
return observer[Pr[ ... ]]
```

Role Convergence Over Time

Simulation: Agents iteratively sample roles, act, and update beliefs



Key Observation: Agents converge to complementary roles without central coordination!

Part 2: Stateful Extensions

Extended Game Mechanics

Health States:

- Both enemy and team have health:
 - DEAD → LOW → MEDIUM → HIGH → FULL
- Win: enemy reaches DEAD
- Lose: team reaches DEAD

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State-Dependent Policies:

- Fighters attack more when enemy is weak
- Tanks defend more when team is vulnerable
- Healers respond to team health levels

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Transition Dynamics:

- Actions affect health state changes
- Probabilistic transitions based on team composition

Player Stats and Prior Beliefs

Player Capabilities (Stats):

Each player has [STR, DEF, SUP]

- P0: [0.7, 0.2, 0.1] → Fighter
- P1: [0.2, 0.7, 0.1] → Tank
- P2: [0.1, 0.2, 0.7] → Healer

Stats determine effectiveness

How to initialize beliefs?

Three types of priors tested

Three Prior Types:

1. Uniform

- All 27 assignments equal
- Learn from scratch

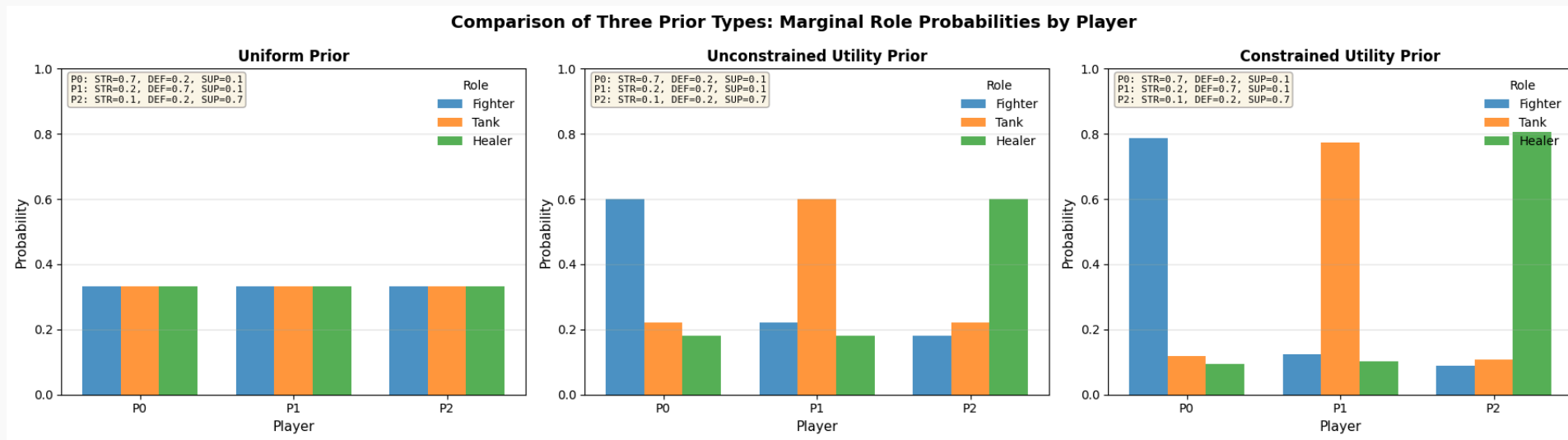
2. Unconstrained Utility

- Prefer stat-matched roles
- Allows duplicates

3. Constrained Utility

- Prefer stat-matched roles
- Enforces unique roles

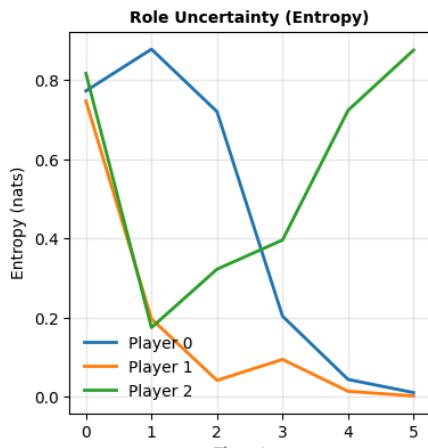
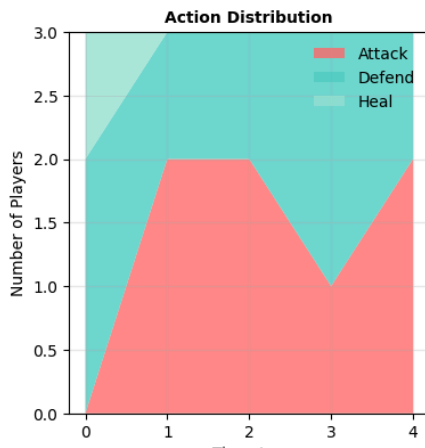
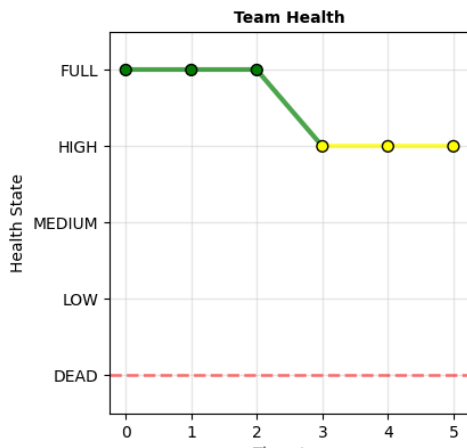
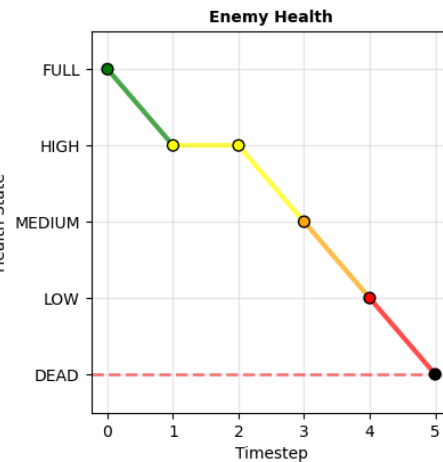
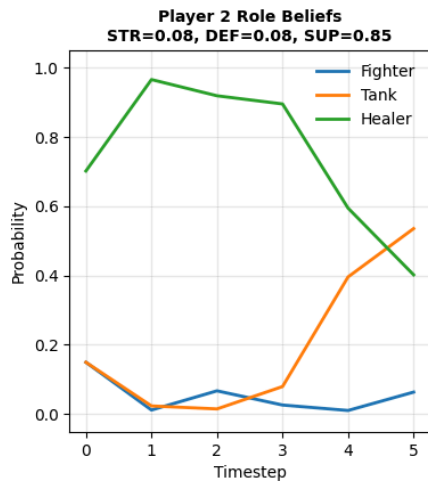
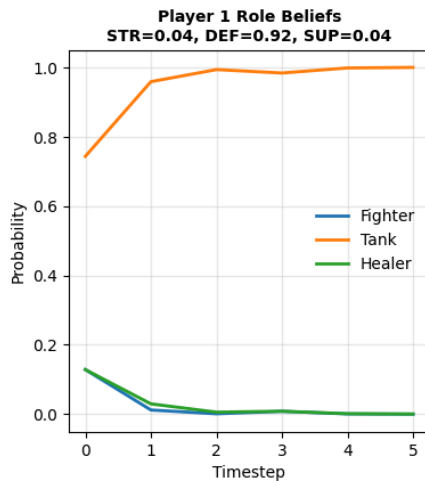
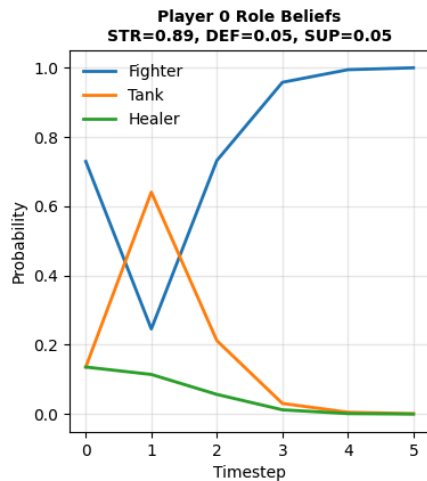
Prior Comparison: Marginal Role Probabilities



Insight: Constrained priors strongly bias toward complementary role assignments

Full Game Dynamics Visualization

Demo Game: Utility-Based Prior



SUMMARY

Outcome: WIN
Duration: 5 steps

Final Roles:
P0: Fighter
P1: Tank
P2: Tank

Actions:
Attack: 7/15 (46.7%)
Defend: 7/15 (46.7%)
Heal: 1/15 (6.7%)

Stat-Optimal: ['Fighter', 'Tank', 'Healer']
Converged To: ['Fighter', 'Tank', 'Tank']

Shows role beliefs, enemy health, team health, action distribution, entropy, and summary stats

Demo Game Key Observations

From the comprehensive visualization, we observe:

Rapid Convergence: Role beliefs stabilize within 2-3 timesteps

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Outcome: Enemy health reaches DEAD, team survives at HIGH health **High**

Confidence: Entropy drops quickly, indicating certainty about roles **Stat**

Alignment: Final roles match each player's best stat

Part 3: Multi-Factor Experiments

Experimental Design

Full factorial design: $3 \times 3 \times 3 = 27$ conditions

Factor 1: Prior Type

- Uniform
- Unconstrained utility
- Constrained utility

Factor 2: Stat Profile

- Random (Dirichlet)
- Balanced (0.33 each)
- Specialist (0.8+ in one)

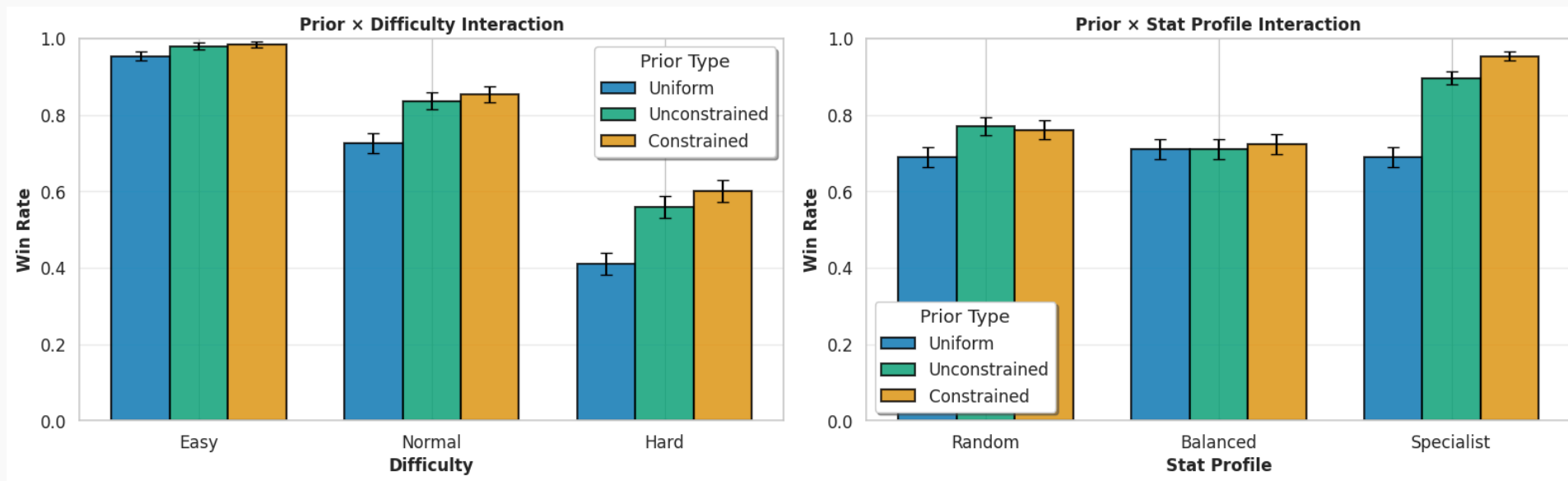
Factor 3: Difficulty

- Easy (0.5×)
- Normal (1.0×)
- Hard (1.5×)

Scale:

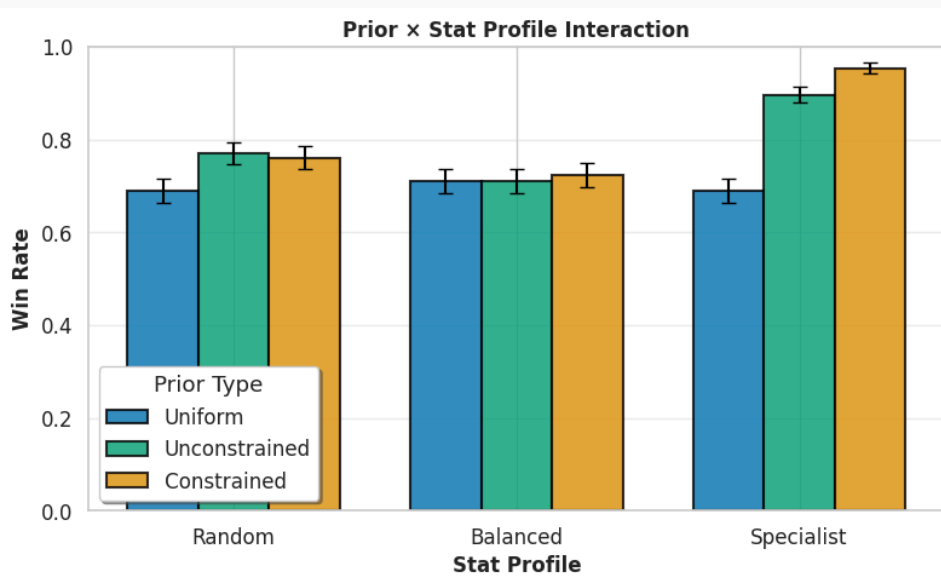
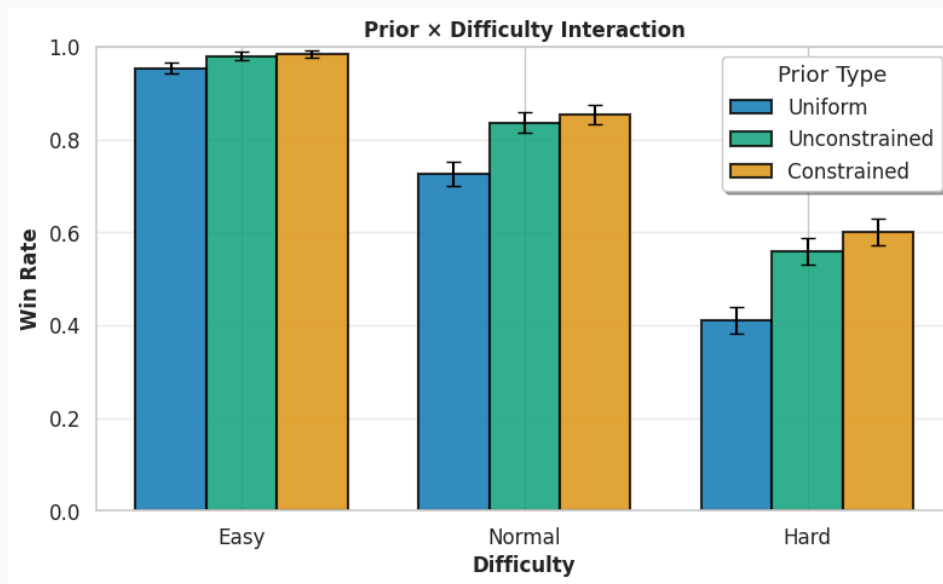
- 100 games per condition
- **Total: 2,700 games**

Main Effects: What Matters Most?



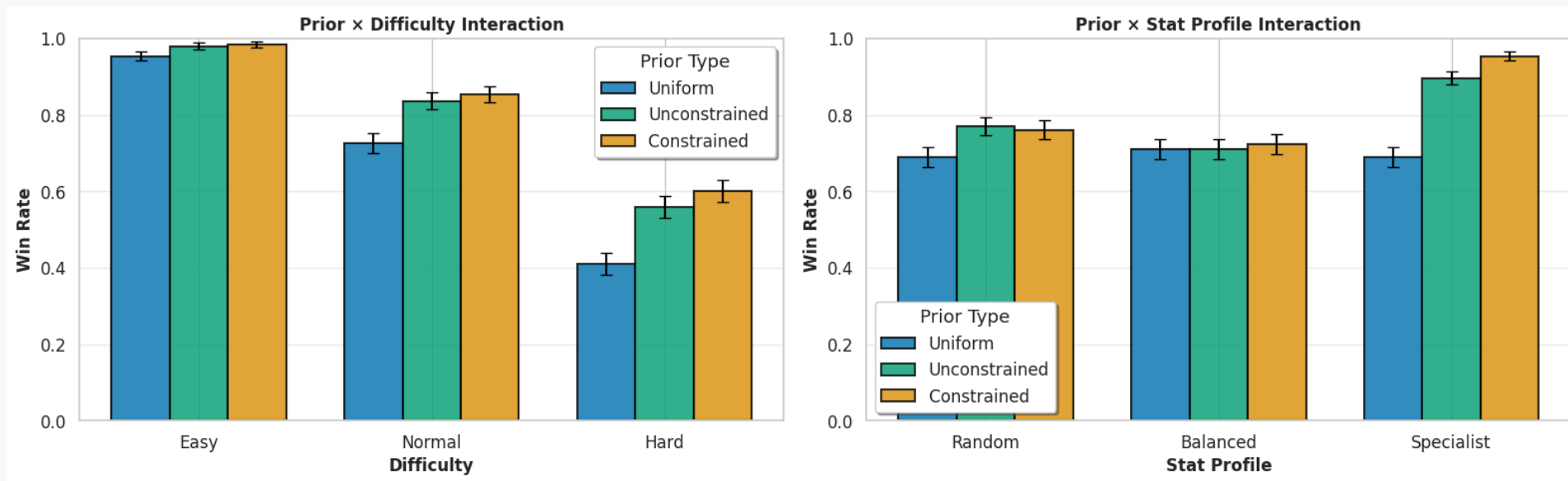
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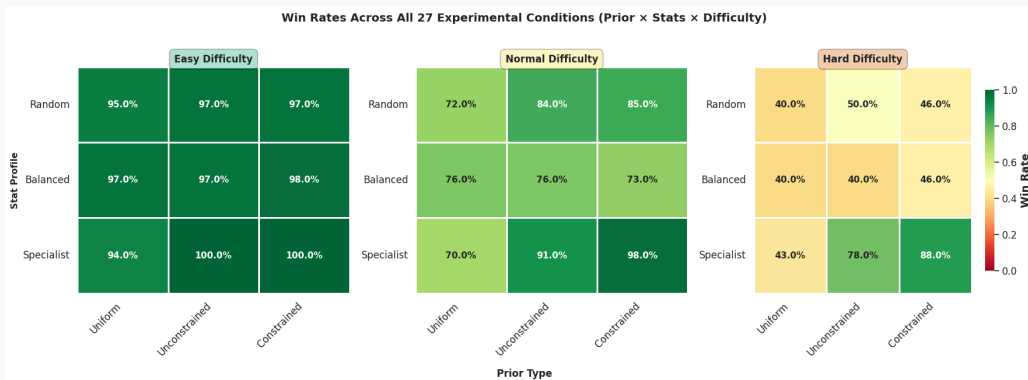
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Stat Profile (Moderate): Specialist (73%) > Balanced (69%) > Random (66%)
Difficulty (Strong): Easy (95%) > Normal (70%) > Hard (40%)

Interaction Effects: Prior × Difficulty × Stats

Key Insights:

Prior × Difficulty:

- Easy: All work (95%)
- Hard: Gap widens
 - Constrained: 83%
 - Uniform: 40%



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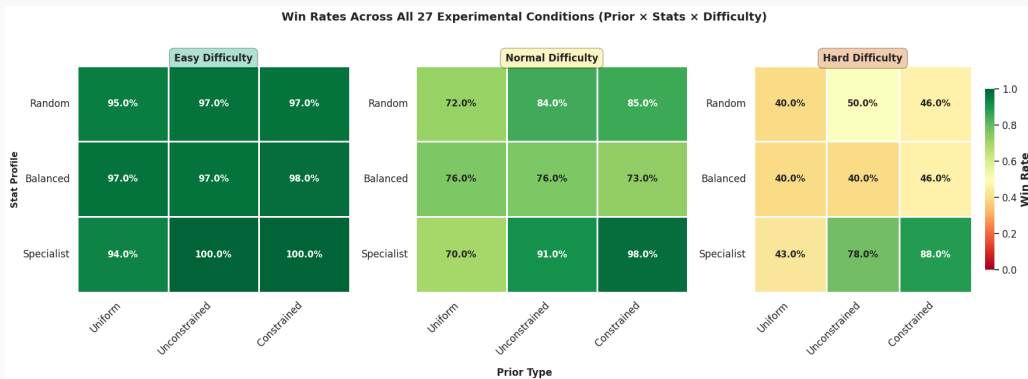
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Prior × Stats:

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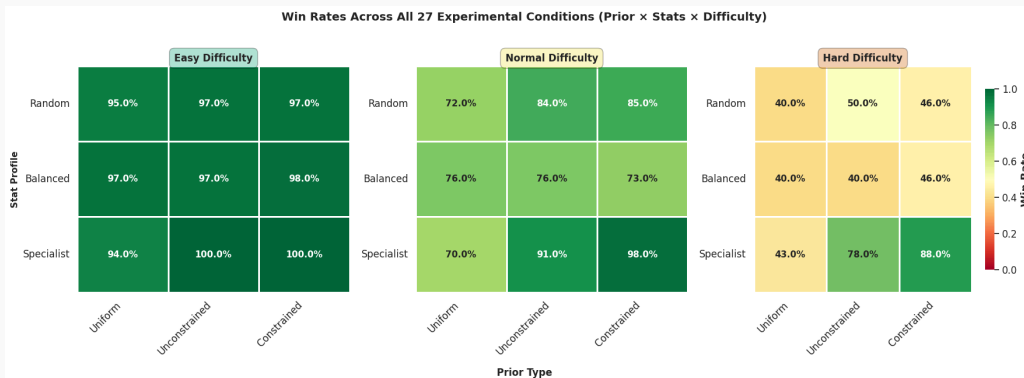
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Takeaway: Value of coordination scales with difficulty

Conclusions

Beyond Win Rate: Coordination Quality

Convergence Speed: Constrained priors converge fastest (fewer timesteps)

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Role Certainty: Lower entropy = more confident beliefs about roles

Pattern: Structured priors + specialized capabilities = superior coordination

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Convergence Speed: Constrained priors converge fastest (fewer timesteps)

Role Certainty: Lower entropy = more confident beliefs about roles **Stat**

Alignment: Specialist teams achieve 67% optimal role matching

Pattern: Structured priors + specialized capabilities = superior coordination

Implications for Cooperative AI Design

Why This Matters:

- Decentralized coordination is fundamental
- Agents must infer structures from behavior
- Bayesian inference handles uncertainty naturally

This work demonstrates how institutional roles emerge

Design Principles:

1. Structured priors

- Encode valid team compositions
- Faster convergence

2. Complementary capabilities

- Heterogeneous > homogeneous
- Clear role matching

3. Test under difficulty

- Easy tasks hide failures
- Hard tasks reveal what works

Key Result:

Constrained utility priors + specialist teams
=
robust coordination even in hard
environments

Summary

What we learned:

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Questions?